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Photonic Nanojet Produced by A Microfluidic Channel for Biofluid Monitoring*

Guoqiang Gu, Pengcheng Zhang, and Hui Yang*

***Abstract*—Owing to the properties like narrow beam waist and high intensity focusing, photonic nanojet (PNJ) has attracted sustainable attention in many fields, such as single nanoparticle or molecular detection, fluorescent or Raman scattering enhancement, subwavelength nanopatterning, super-resolution imaging, etc. The characteristics of PNJ are mainly affected by the material compositions and morphology features of the investigated microstructures. In this paper, we present a microfluidic channel structure to produce PNJs for biofluid monitoring. The proposed scheme is numerically studied by using the finite element method. Through the change of PNJ parameters, such as the maximum light intensity, working distance and full width at half-maximum, the refractive indices of the injected biofluids can be determined.**

I. BACKGROUND

Biofluid monitoring plays an important role in the field of life science. The physical properties of biofluid such as refractive index, density, viscosity, conductivity and opacity can provide abundant information for monitoring [1]. Refractive index is a very interesting property as it contains a lot of important formation on such as blood glucose level and label-free molecular interaction [2, 3]. Optical measurement on the refractive index has attracted wide attentions owning to its reliability, fast response and high sensitivity [4]. Photonic nanojet (PNJ), a scattered beam of light with a high-intensity main lobe and very narrow waist that can be generated upon illumination on a transparent dielectric symmetric microstructure. Therefore, PNJ is a good candidate for monitoring the refractive index of liquid [5, 6], as the field distributions and shape features of the PNJ are mainly determined by the geometrical morphologies and material properties of the microstructures [7-11]. The experimental characterization of PNJ can be performed using the method proposed in Ref. [12].

Here, we design a microfluidic channel structure consisted of a glass tube and filled with biofluids. The microfluidic structure is designed with geometry specific for the generation of PNJ. The characteristics of the PNJ are theoretically investigated by numerical simulation using finite element method (FEM). The parameters of the PNJ, i.e. maximum light intensity I_{max}, working distance (WD) and full-width at half maximum (FWHM) changing with the refractive indices of different biofluids filled in the microfluidic channel, are analyzed. The formed two different PNJ modes based on the inflection point of the first light refraction interface potentially provide more ways for probing. Our research will facilitate the study of integrating PNJ phenomenon with microfluidic devices as well as the exploitation of new applications based on such integrated system.

II. RESEARCH APPROACH

The schematic of the research model is shown in Fig.1(a). The microfluidic channel made of glass consists of one hollow microtubule in the middle and two cuboids on both sides. Biofluids can be injected through the inlet channel of the cuboid on the left, filling the hollow microtubule and flowing out of the cuboid on the right. The illumination light is incident on the top of the microfluidic

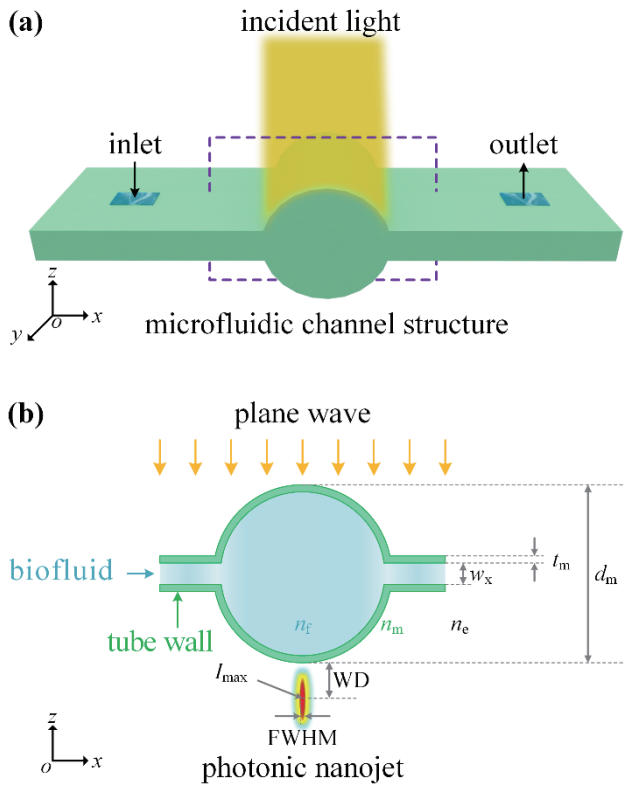

Fig. 1. Schematic diagrams of plane-wave-illuminated microfluidic channel system. (a) Three-dimensional (3D) structural model and (b) two-dimensional (2D) cross-section showing the simulation model.

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All authors are with the Laboratory of Biomedical Microsystems and Nano Devices, Bionic Sensing and Intelligence Center, Institute of Biomedical and Health Engineering, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, 518055 Shenzhen, China (phone: +86-755-8639-2675; e-mail: hui.yang@siat.ac.cn).

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channel system. The model is assumed infinite long along y-axis direction.

Fig. 1(b) shows the cross-section of the region marked by the dotted box in Fig. 1(a), corresponding to the 2D model used in the following numerical studies. All the results are obtained from the 2D FEM-based full-wave simulation. The unpolarized plane wave with monochromatic wavelength (λ) and unit light intensity (I) is propagating from the top boundary to the bottom boundary of the computational domain along negative z-axis direction. n_b, n_m and n_e are the refractive indices of the filled-biofluid, the microfluidic channel and the environmental medium, respectively. d_m is the diameter of the middle cylinder, t_m is the tube wall thickness of the microfluidic channel, w_x is the width of the inlet along x-axis direction. Three main PNJ parameters are discussed in this work: I_{max} is the maximum light intensity in the axial profile, WD is the distance from the vertex at the bottom of the middle cylinder to the point of I_{max}, FWHM is the beam waist of the central diffraction lobe along x-axis direction. The origin of the coordinate axis is defined at the center of the microfluidic channel system.

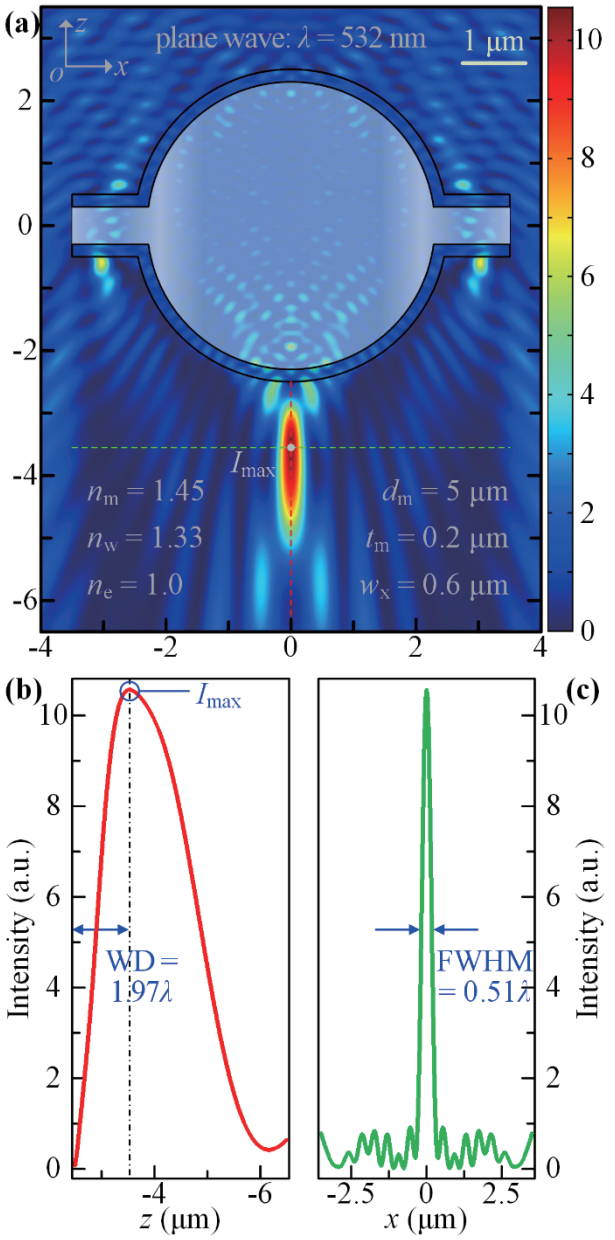

Fig. 2. (a) FEM simulation results for a typical example of PNJ generated by illuminating a microfluidic channel system with $n_m = 1.45$, $n_w = 1.33$, $n_e = 1$, $d_m = 5$ μm, $t_m = 0.2$ μm, $w_x = 0.6$ μm, and $\lambda = 532$ nm, $I = 1$. (b) Light intensity profile along z-axis (from $z = -2.5$ μm to $z = -6.5$ μm). (c) Lateral light intensity at the I_{max} point along x-axis direction.

III. Results And Discussion

Fig. 2(a) shows the light intensity distribution of a water flow filled microfluidic channel that is illuminated by a plane wave. The wavelength of the monochromatic wave is $\lambda = 532$ nm, and the intensity is $I = 1$. The refractive indices of the glass tube, filled water and environmental medium are $n_m = 1.45$, $n_w = 1.33$ and $n_e = 1$, respectively. Here, the geometric features of the investigated model are $d_m = 5$ μm, $t_m = 0.2$ μm, $w_x = 0.6$ μm, respectively. A typical PNJ is formed on the rear side of the middle microcylinder. The intensity distribution along the line between the point $x = 0$ μm, $z = -2.5$ μm and the point $x = 0$ μm, $z = -6.5$ μm, which is marked with red dashed line in Fig. 2(a), is shown in Fig. 2(b). The maximum light intensity I_{max} is located at the point of $x = 0$ μm, $z = -3.55$ μm. The value of I_{max} is more than ten times higher than the intensity of the incident plane wave. The working distance WD equals to 1.97λ, which means the focal point position is beyond the decay length of the evanescent wave ($\lambda/2\pi$ [7]). At the I_{max} point along x-axis direction, the intensity profile is drawn in Fig. 2(c). By fitting the distribution curve with a Gaussian approximation, the obtained FWHM is ~0.51λ. It is within the subwavelength scale and close to the classical diffraction limit.

A distinct advantage of the microfluidic channel structure presented in this work is the sensitivity of PNJ parameters to the filled fluidic media. Fig. 3 shows the correlation between PNJ parameters and the refractive indices of different biofluids. The dielectric properties of the

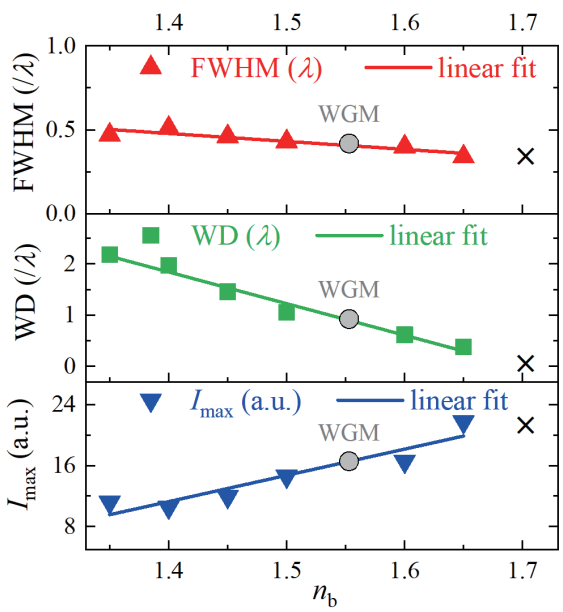

Fig. 3. PNJ parameters (I_{max}, WD and FWHM) for the injected biofluids with different refractive indices. The gray circles and black crosses represent the possible WGMs and the near-field focusing (not PNJ).

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microfluidic channel and the surrounding medium, as well as the dimensional parameters of the microchannel keep the same as the parameters given in Fig. 2(a). As the refractive indices of the biofluids to be measured increase from 1.35 to 1.65 with step size of 0.05, the I_{max} shows an increasing trend, while the WD and FWHM both have a decreasing trend. All the lateral beamwidths are equal to or below the diffraction limit. The cases of $n_b = 1.55$ and $n_b = 1.70$, which are marked with circle and cross symbol in Fig. 3, represent the special situation of the possible resonant whispering gallery modes (WGMs) and not the normally defined PNJ, respectively.

As shown in Fig. 4(a), the distribution of the optical field around the interface between the filled biofluid and the tube is similar to the resonant WGMs [13]. Due to the generation of WGMs, the photon is confined within the microfluidic channel and travels back and forth, leading to the I_{max} point moving toward the inside of the microfluidic channel. In transverse electric (TE) mode, we calculated the resonant modes with radial quantum number $n = 1$, angular quantum number $l = 27$ [14]. Fig. 4(b) and Fig. 4(c) are the mode field distributions in the equator and polar directions. The resonant wavelength of the fundamental mode for the microfluidic channel cavity is $\lambda \approx 531.32$ nm, which is very close to the illumination wavelength. Because the refractive index difference between the filled biofluid of $n_b = 1.55$ and the glass tube of $n_m = 1.45$ is very small, the ability of microfluidic channel to confine the photons is weak and thus the weak light fields existed in the periphery of the cavity. When the refractive index of the filling biofluid further increases to 1.70, which is marked with the cross symbol in Fig. 3, the I_{max} point returns to the interior of the microfluidic channel. As can be seen from Fig. 4(d), the propagation light

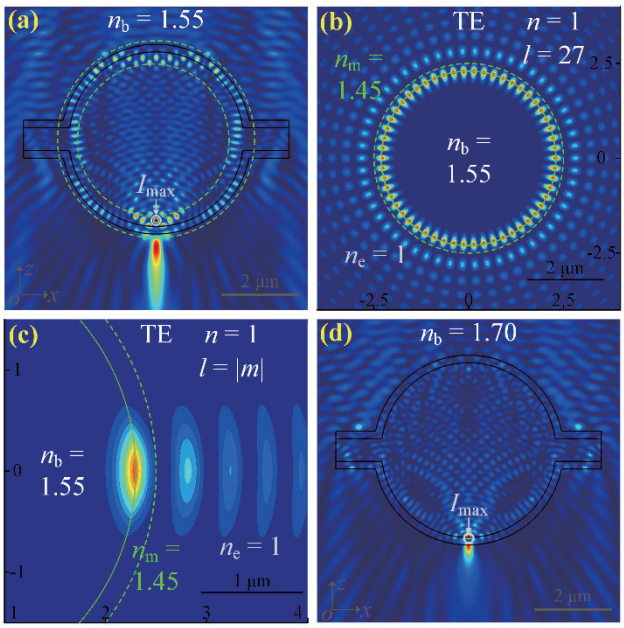

Fig. 4. (a) optical field distribution of the microfluidic channel with the refractive index of filled biofluid is $n_b = 1.55$. The green lines marked field distribution is the possible existed resonant WGMs. The TE mode field distributions of the microfluidic channel cavity along (b) equal direction with $n = 1$, $l = 27$ and (c) polar direction with $n = 1$, $l = 27$, $l = |m|$. m is the azimuthal quantum number. (d) optical field distribution of the microfluidic channel with the refractive index of filled biofluid is $n_b = 1.70$.

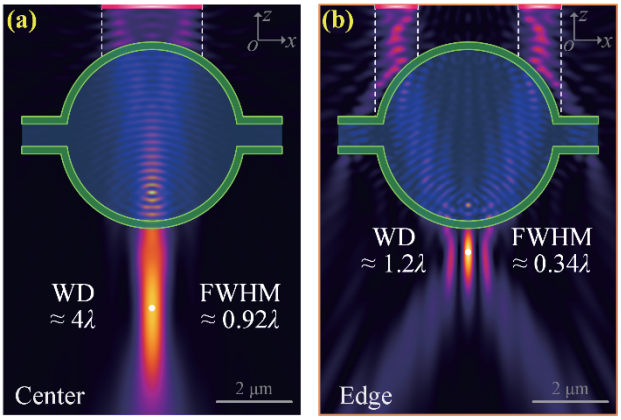

Fig. 5. Two different types of PNJs with (a) long WD, wide FWHM and (b) short WD, narrow FWHM used for biofluids monitoring that is generated from the middle region and two edge regions illuminated by the plane wave.

is focused within the microfluidic channel system and there exists no PNJ.

There is a universal rule for plane-wave-illuminated circular-structured dielectric microparticles: when the two edge regions of the first light refraction interface are illuminated, a tight focus with small FWHM and short WD is formed; on the other hand, when the middle region of the first light refraction interface is illuminated, a loose focus with large FWHM and long WD is formed. The inflection point, i.e. the boundary point between the edge and the middle regions, can be solved by a specific mathematical and physical method [15]. Based on the knowledge of the inflection point, we calculate the optical field distributions of the microfluidic channel structure under two different irradiation conditions. Since the circular boundary part of the microfluidic channel structure plays a major role in the formation of PNJ, we set the irradiation light to only illuminate the circular boundary part of the designed structure. The coordinates of the calculated two inflection points are (-1.36 μm, 2.1 μm) and (1.36 μm, 2.1 μm), respectively. Fig. 5(a) and Fig. 5(b) show the light field distributions of the abovementioned two cases. It is clear to see that the PNJs for these two cases are different. When the middle region is illuminated, the formed PNJ has a long WD (~ 4λ) and a wide FWHM (~ 0.92λ) without any secondary diffraction side lobes. While the two edge regions are illuminated, the produced PNJ has one central main lobe and two secondary side lobes. The WD and FWHM of the main lobe is ~ 1.2λ and ~ 0.34λ, respectively. Obviously, such two different types of PNJs will potentially provide more approaches for the monitoring biofluids.

IV. Conclusion

In summary, we designed and demonstrated a microfluidic channel structure for the generation of PNJs. The model is preliminarily investigated by numerical calculation with FEM-based simulation and theoretical analysis with the method of mathematical physics. The formed PNJ has a beam waist close to the diffraction limit, a maximum light intensity an order of magnitude higher than the intensity of the illumination source and a propagation distance far beyond the evanescent field region

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without significant diffraction. Through the variation of the PNJ parameters, the injected biofluids into the microfluidic channel with different refractive indices can be effectively perceived and monitored. The WGMs appear when the injected biofluid causes the captured photon to satisfy the resonant condition in the microfluidic channel cavity. The PNJs of different light field structures from the division of middle and edge irradiated regions based on the inflection point indicate more approaches for refractive index detection. Compared with other monitoring and detection methods, this approach provides advantages such as straightforwardness, high throughput and high sensitivity, and the possibility of multiplex detection. The proposed technique here will not only provide new approaches to integrate the PNJs into microfluidic devices, but also open new opportunities for developing new PNJ-based techniques for microfluidic biosensing applications.

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